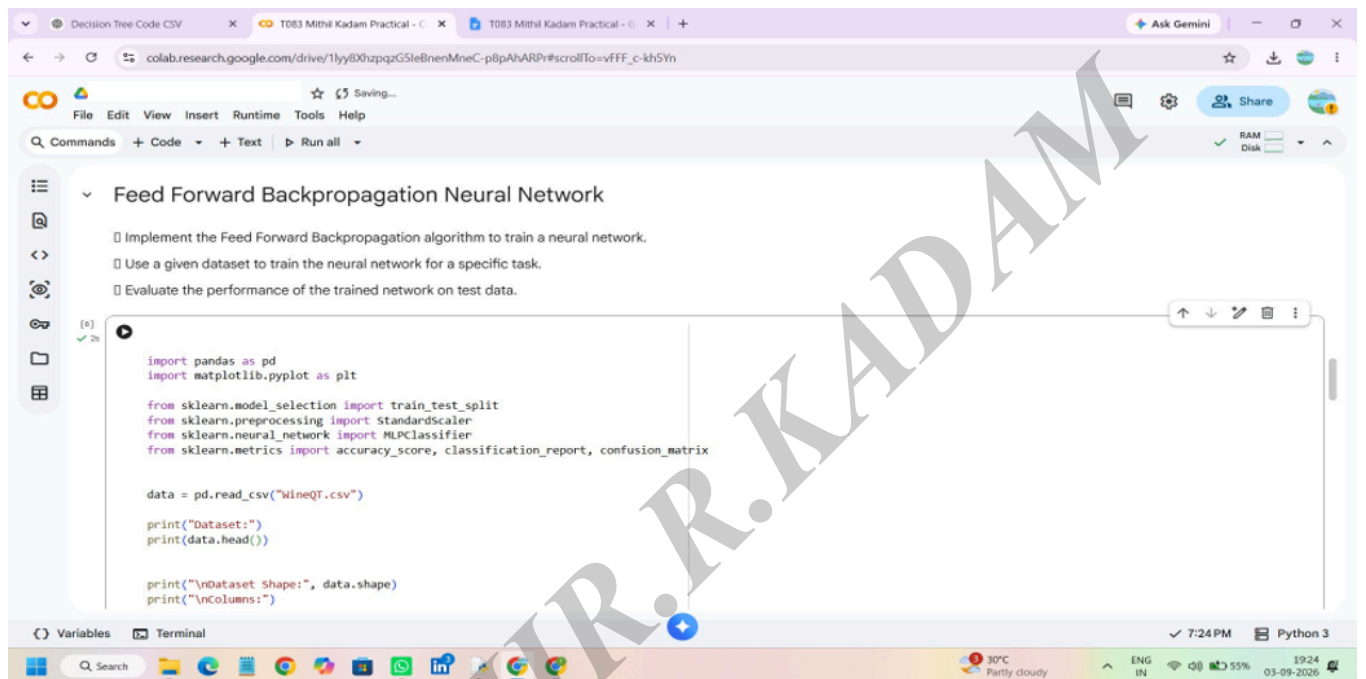


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Aim:- Feed Forward Backpropagation Neural Network

Implement the Feed Forward Backpropagation algorithm to train a neural network.

Use a given dataset to train the neural network for a specific task. Evaluate the performance of the trained network on test data.



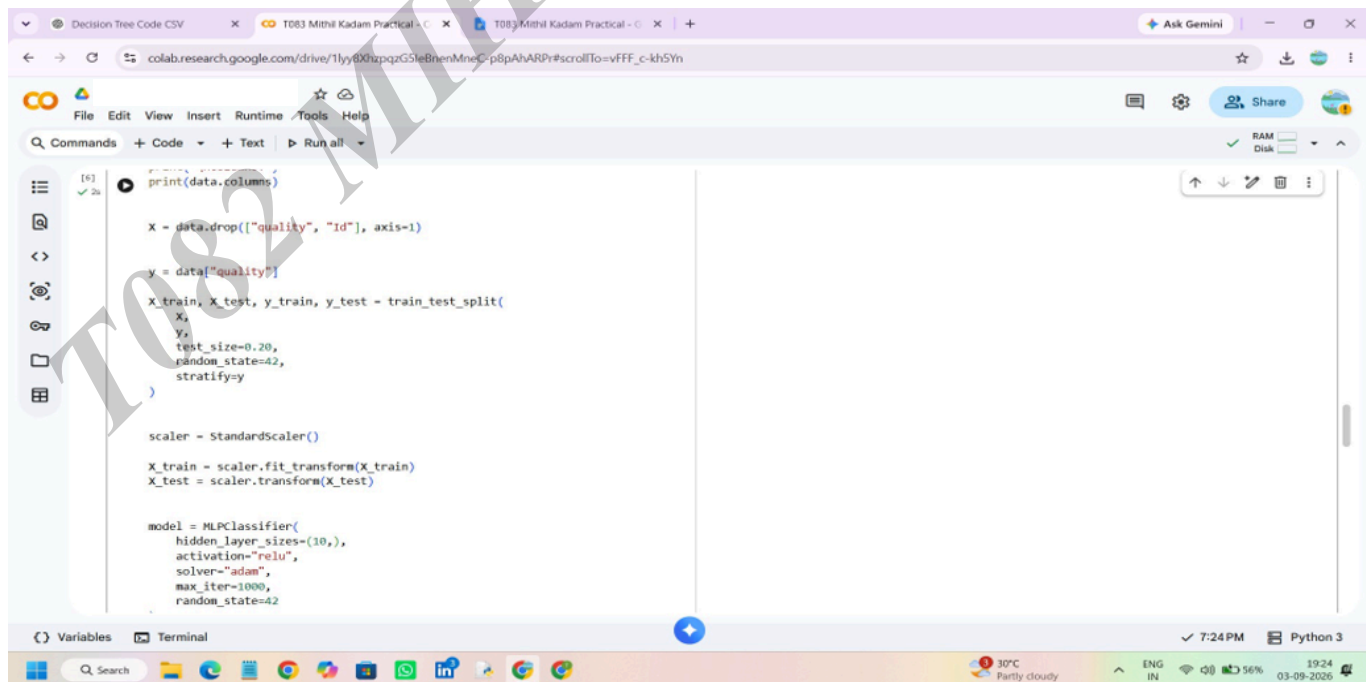
```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

data = pd.read_csv("wineQT.csv")

print("Dataset:")
print(data.head())

print("\nDataset Shape:", data.shape)
print("\nColumns:")
```



```
print(data.columns)

X = data.drop(["quality", "Id"], axis=1)
y = data["quality"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

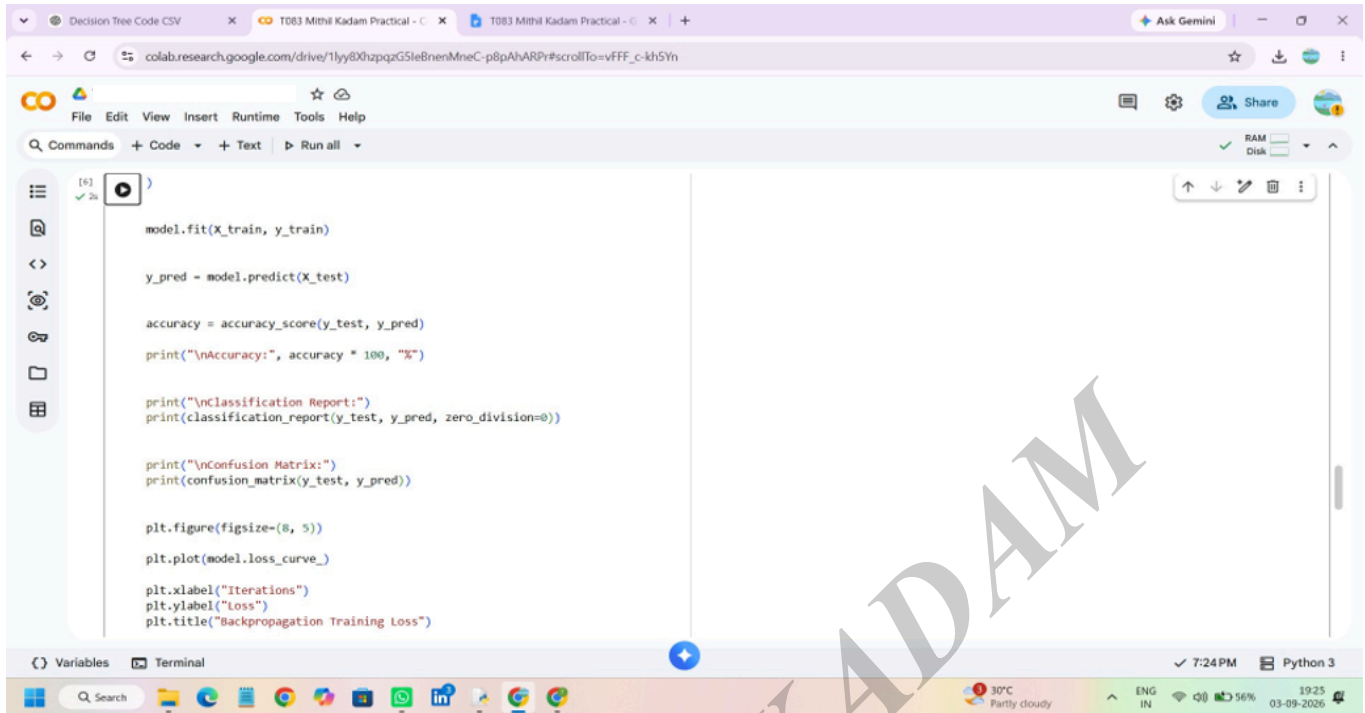
model = MLPClassifier(
    hidden_layer_sizes=(10,),
    activation="relu",
    solver="adam",
    max_iter=1000,
    random_state=42
)
```

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```
[6] ✓ 2s )
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy:", accuracy * 100, "%")

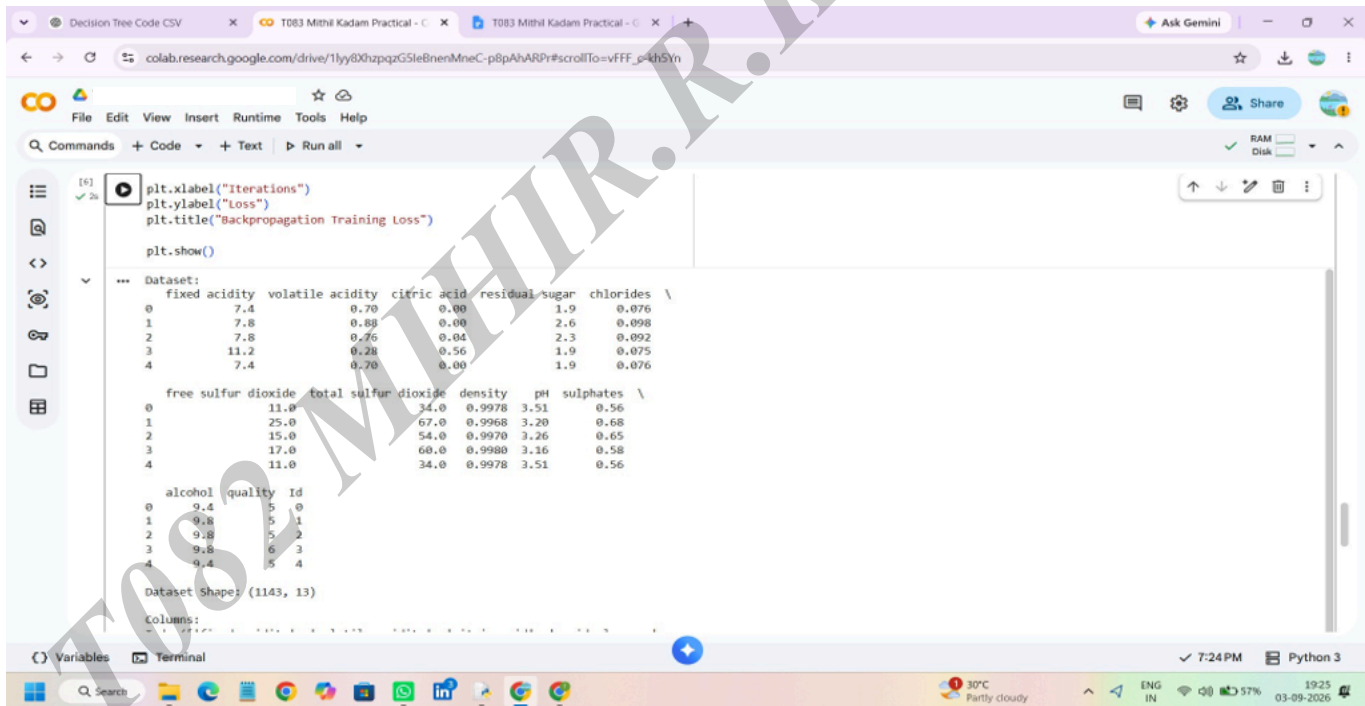
print("\nClassification Report:")
print(classification_report(y_test, y_pred, zero_division=0))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

plt.figure(figsize=(8, 5))

plt.plot(model.loss_curve_)

plt.xlabel("Iterations")
plt.ylabel("Loss")
plt.title("Backpropagation Training Loss")
```



```
[6] ✓ 2s )
plt.xlabel("Iterations")
plt.ylabel("Loss")
plt.title("Backpropagation Training Loss")
plt.show()
```

Dataset:

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides
0	7.4	0.70	0.00	1.9	0.076
1	7.8	0.88	0.00	2.6	0.098
2	7.8	0.76	0.04	2.3	0.092
3	11.2	0.28	0.56	1.9	0.075
4	7.4	0.70	0.00	1.9	0.076

	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates
0	11.0	34.0	0.9978	3.51	0.56
1	25.0	67.0	0.9968	3.20	0.68
2	15.0	54.0	0.9970	3.26	0.65
3	17.0	60.0	0.9980	3.16	0.58
4	11.0	34.0	0.9978	3.51	0.56

	alcohol	quality	Id
0	9.4	5	0
1	9.8	5	1
2	9.8	5	2
3	9.8	6	3
4	9.4	5	4

Dataset Shapes (1143, 13)

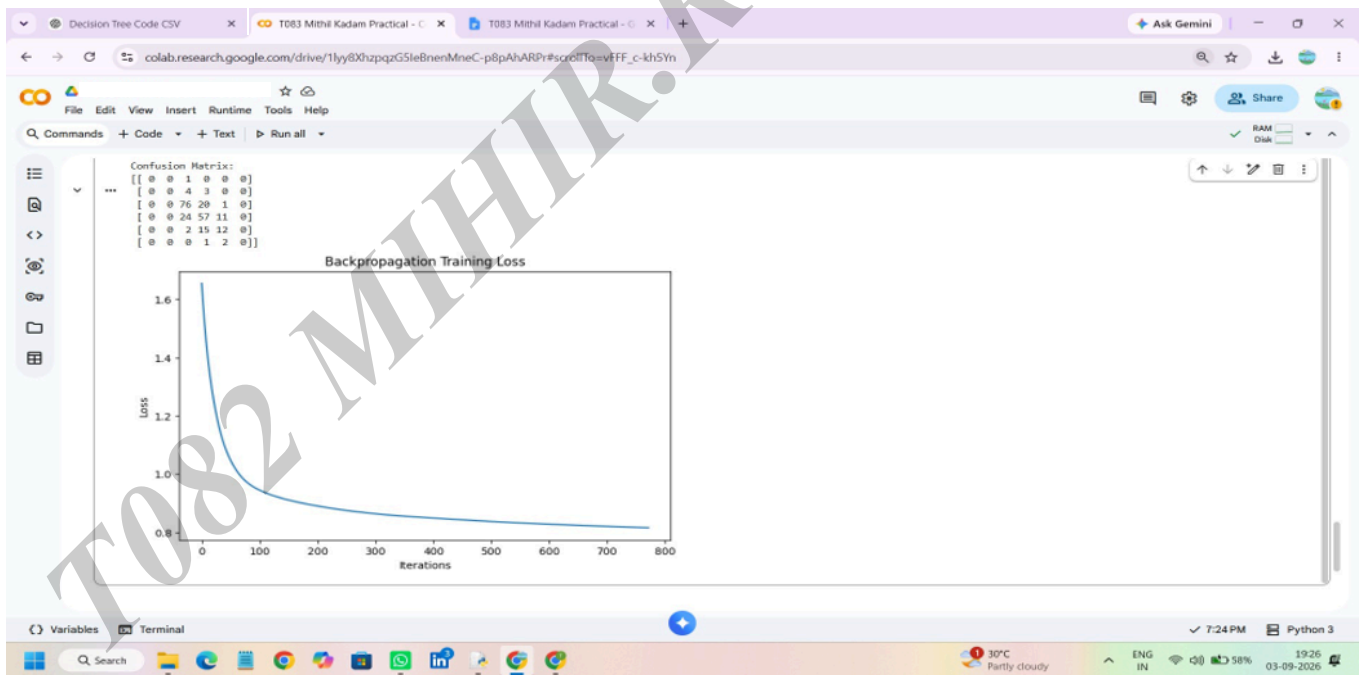
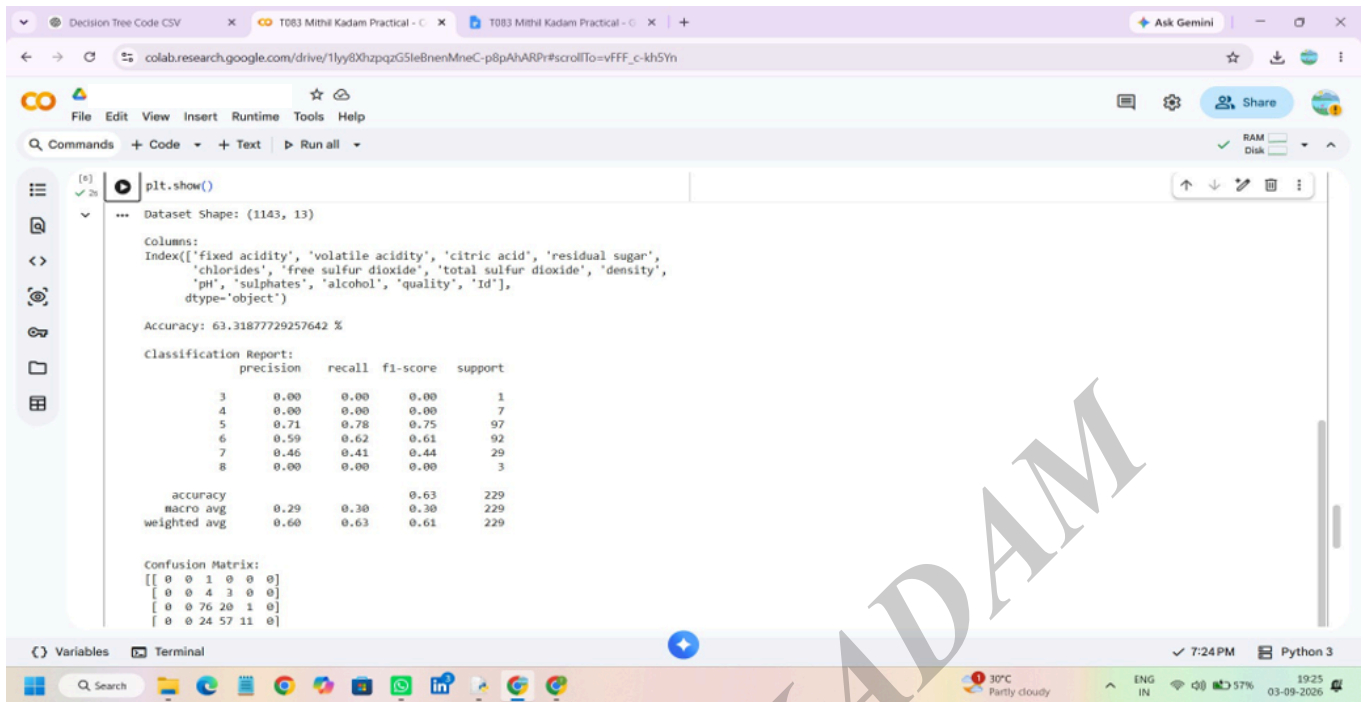
Columns:

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